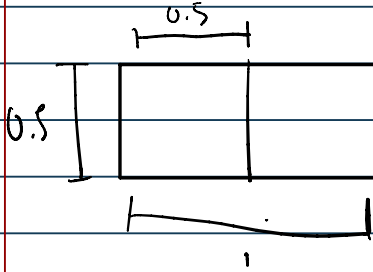


Homework 7 CHE 312 Katrina Mulcahy

Monday, March 2, 2020

11:13 AM

7.17



$$T_s = 10^\circ\text{C}$$

$$T_\infty = 35^\circ\text{C}$$

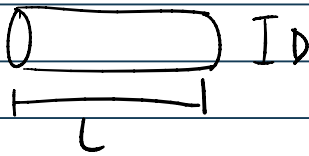
$$u_\infty = 25 \text{ m/s}$$

20

Having the rough part be in the front would minimize the convection heat transfer since the air is hotter than the surface

5

7.38



$$D = 0.1 \text{ m}$$

$$L = 2 \text{ m}$$

$$T_s = 70^\circ\text{C}$$

$$V = 3 \text{ m/s}$$

$$T_\infty = 20^\circ\text{C}$$

$$\approx 300 \text{ K}$$

air at 300 K

$$\rho = 1.1614 \text{ kg/m}^3$$

$$\nu = 15.89 \cdot 10^{-6} \text{ m}^2/\text{s}$$

$$k = 26.3 \cdot 10^{-3} \text{ W/mK}$$

$$c_p = 1.007 \text{ kJ/kgK}$$

$$\mu = 28.97 \text{ kg/kmol}$$

15

$$Re = \frac{V \cdot D}{\nu} = \frac{3 \frac{\text{m}}{\text{s}} \cdot 0.1 \text{ m}}{15.89 \cdot 10^{-6} \text{ m}^2/\text{s}} = 1.89 \cdot 10^8$$

$$Pr = \frac{c_p \mu}{k} = \frac{1.007 \cdot 28.97}{26.3 \cdot 10^{-3}} = 0.00111$$

$$Nu = 0.023 (Re)^{0.8} (Pr)^{0.3}$$

$$\begin{aligned}
 Nu_f &= 0.023 (Re)^{0.8} (Pr)^{0.3} \\
 &= 0.023 (1.89 \cdot 10^{-8})^{0.8} (0.00111)^{0.3} \\
 &= 1.98 \cdot 10^{-9}
 \end{aligned}$$

$$R = \frac{1}{hA} = \frac{1}{1.98 \cdot 10^{-9} \cdot 2 \cdot \pi \cdot 0.1}$$

$$R = 8.03 \cdot 10^9 \text{ K/m}$$

$$\frac{\Delta T}{R} = \frac{10-20}{8.03 \cdot 10^9} = 1.24 \cdot 10^{-9}$$

	A	B	C	D	E	F	G	H	I	J
1	Diameter	Re	h							
2	0.05	9.4399E-09	1.13556E-09							
3	0.1	1.888E-08	1.97712E-09							
4	0.15	2.832E-08	2.73467E-09							
5	0.2	3.776E-08	3.44236E-09							
6	0.25	4.7199E-08	4.11514E-09							
7	0.3	5.6639E-08	4.76134E-09							
8	0.35	6.6079E-08	5.38625E-09							
9	0.4	7.5519E-08	5.9935E-09							
10	0.45	8.4959E-08	6.58571E-09							
11	0.5	9.4399E-08	7.16487E-09							
12										
13	Pr									
14	0.001109									
15										
16										
17										

